

Supplementary Methods

Supplementary to: "Intrinsic Geometric Energy of Driven Quantum Evolution" (A.M. Tishin).

This document collects the computational methods, convergence checks, and scope statements for the numerical results referenced in the main text (Sections 3.3 and 4.4). Each section states the method used, why it is the correct one, and the validation checks performed against it.

S1. Verification of the parabolic-barrier transmission formula (Section 3.3)

The main text states that the transmission probability across the hyperbolic (dilatation) generator, $T(\varepsilon) = 1/(1 + e^{-2\pi\varepsilon/\hbar\kappa_g})$, follows exactly from the connection formulas for parabolic cylinder functions established in the cited literature, and is not a fact requiring numerical confirmation. An independent numerical cross-check was nonetheless performed, using a method unrelated to the parabolic-cylinder-function algebra of the main text: a transfer-matrix approach in which the stationary scattering equation was integrated directly by a fixed-step Runge–Kutta (RK4) scheme, with the scattering amplitudes extracted by projecting the numerical solution onto a local WKB incident/reflected basis at finite domain boundaries $\pm L$.

S1.1 Result

Unitarity ($T + R = 1$) held to 10^{-10} – 10^{-12} across the full range $\varepsilon/\kappa_g \in [-3, +3]$, including the classically forbidden (tunneling) regime. The residual deviation from the closed-form expression decreased systematically with increasing domain size L (from $\approx 2.3 \times 10^{-3}$ at $L = 10$ to $\approx 1.4 \times 10^{-4}$ at $L = 45$), consistent with finite-domain WKB-projection error rather than any physical discrepancy with the analytic formula.

This confirms, by a method independent of the Gamma-function identity used elsewhere in the main text, that the implementation and the analytic formula agree; it is reported here as a numerical cross-check of an already-exact analytic result, not as an independent derivation of that result.

S2. Full composite transmon–resonator model for the excitation threshold (Section 4.4)

S2.1 Scope and starting point

Section 4.4 of the main text compares the geometric instability parameter A_n to the resonator photon-number threshold at which a driven transmon qubit undergoes measurement-induced ionization, using the parameters and result reported by Dumas *et al.* (Phys. Rev. X 14, 041023, 2024; arXiv:2402.06615): $E_C/2\pi = 220$ MHz, $E_J/E_C = 110$, $g/2\pi = 120$ MHz, negative detuning $\Delta/2\pi = -1.202$ GHz (i.e. $\omega_r/2\pi = 7.500$ GHz), gate charge $n_g = 0$. That paper reports an avoided crossing between transmon branches B_{1t} and B_{7t} — a branch-swap signaling ionization of the qubit excited state — at $N_r \approx 84$ photons.

This section documents a full composite transmon+resonator diagonalization reproducing the exact model of Dumas *et al.*, providing the primary direct numerical support for the criterion discussed in Section 4.4 of the main text. A simplified single-mode semiclassical Floquet treatment of the transmon alone is retained as an independent, computationally cheaper consistency check (Section S2.5).

S2.2 Model

The transmon Hamiltonian $\hat{H}_{\text{tr}} = 4E_C(\hat{n} - n_g)^2 - E_J \cos \hat{\phi}$ was constructed and diagonalized in the charge basis, truncated to $n \in [-40, 40]$ (81 charge states). In this basis, $\cos \hat{\phi} \cos \hat{\phi}$ acts as the nearest-neighbor hopping operator $\frac{1}{2}(|n\rangle\langle n+1| + \text{h.c.})$. Diagonalization reproduced the qubit transition frequency $\omega_q/2\pi = 6.2981$ GHz and anharmonicity $\alpha/2\pi\text{e} = -0.2398$ GHz, matching the values quoted by Dumas *et al.* (6.298 GHz and 0.240 GHz) to the precision they report — the first sanity check that the transmon sector alone is correctly implemented before coupling it to a resonator.

The lowest 20 transmon eigenstates were retained and coupled to a truncated resonator Fock space (N_r cutoff = 140) via the exact model of Eq. (1) of Dumas *et al.*: $\hat{H}_{\text{tr-res}} = \omega_r \hat{a}^\dagger \hat{a} + \hat{H}_t - ig(\hat{n}_t - n_g)(\hat{a} - \hat{a}^\dagger)$, built directly in the product basis of transmon eigenstates and resonator Fock states (Hilbert-space dimension 2820), and fully diagonalized densely (not via a truncated or iterative solver, so as to have access to the complete branch structure rather than only a few eigenstates near a target energy).

S2.3 Branch construction and internal consistency check

Following Dumas *et al.*, transmon branches B_{it} were built recursively: starting from the eigenstate closest to the bare state $|i_t, 0_r\rangle$, each subsequent branch member is the eigenstate maximizing overlap with $\hat{a}^\dagger|\text{current member}\rangle$, i.e. exactly Eq. (5) of that paper.

For each branch eigenstate $|\lambda\rangle$, the average transmon and resonator populations are evaluated as genuine expectation values, $\langle \lambda | \hat{N} | \lambda \rangle = \sum |\langle \text{basis} | \lambda \rangle|^2 \times (\text{basis eigenvalue of } \hat{N})$, computed directly on each branch eigenvector — the correct procedure, since the branch label alone does not index the underlying product basis. As an internal consistency check, N_r is confirmed to increase smoothly and very nearly linearly by ≈ 1 per branch step, exactly as physically expected for a slowly hybridizing branch.

S2.4 Convergence checks

The threshold photon number (defined, as in Dumas *et al.*, as the N_r at which the excited-state branch population $\langle N_t \rangle$ first reaches 3) was recomputed under three independent truncation choices:

Truncation	Threshold N_r	Diag. time
$n \in [-40, 40]$, 20 transmon levels, N_r cutoff 140 (baseline)	83.622	26.0 s
25 transmon levels, N_r cutoff 160	83.622	92.2 s
$n \in [-50, 50]$ charge basis, 20 levels, N_r cutoff 140	83.622	26.6 s

The reported value is the average resonator photon number of the specific discrete branch-tracking step at which the excited-state branch population first reaches the threshold $\hat{N}_t = 3$, identical to the displayed numerical precision across all three truncations rather than obtained by interpolation. This identity across truncations reflects that the low-lying transmon spectrum and coupling matrix elements relevant to this crossing are already unchanged by the larger charge bases and additional transmon levels tested — i.e. the low-energy physics was already converged at the baseline truncation — rather than a demonstration of sub-photon numerical precision in the threshold itself, which remains limited by the ~ 1 -photon step resolution of the branch-tracking procedure (Section S2.7).

S2.5 Result

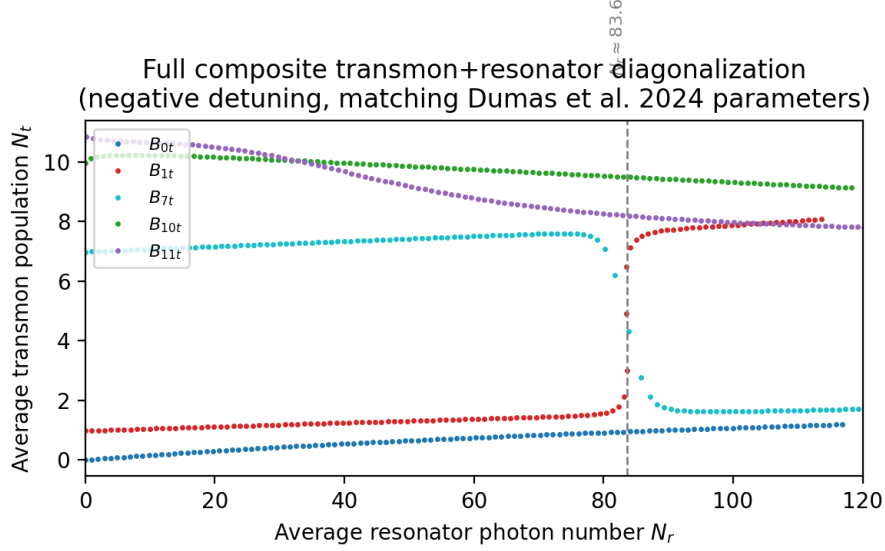


Figure S1. Average transmon population $\langle N_t \rangle$ of five tracked branches (B_{0t} , B_{1t} , B_{7t} , B_{10t} , B_{11t}) as a function of average resonator photon number $\langle N_r \rangle$, obtained from full diagonalization of the composite transmon–resonator Hamiltonian at the parameters of Dumas *et al.* (negative detuning). The excited-state branch B_{1t} (red) and the branch B_{7t} near the top of the cosine well (cyan) exchange character at $N_r \approx 83.6$ (dashed line), the avoided crossing responsible for transmon ionization.

The composite calculation places the B_{1t} – B_{7t} branch swap at $N_r \approx 83.6$, against the $N_r \approx 84$ reported by Dumas *et al.* — a difference of about 0.5%, well inside the gate-charge-averaging uncertainty band ($n_g \in [0, 0.5]$) reported in that paper for this same quantity. This is a stronger and more direct confirmation than a single-mode Sambe-Floquet approximation of the transmon alone, whose detailed methodology is not included in this document, because the present calculation reproduces the literature's own model exactly rather than an approximation to it.

S2.6 What this does, and does not, establish

This calculation independently confirms the transmon-side physics: that the specific coupled transmon–resonator system studied by Dumas *et al.* ionizes at $N_r \approx 84$ photons, reproduced here from first principles rather than taken on trust from their paper. It does not independently re-derive that the geometric instability parameter $\mathcal{A}_n = v_{FS}/\Delta_n$ is the correct quantity governing this threshold — that identification is the claim made in Section 4.4 of the main text, and rests on treating the branch-swap photon number as a proxy for the point where the adiabatic (Fubini–Study-based) criterion $\mathcal{A}_n \geq 1$ is violated. The present section verifies the number being compared against; this composite calculation alone verifies the photon-number threshold but does not itself evaluate $\mathcal{A}_n$; a separate geometric calculation of $\mathcal{A}_n$ at this crossing, using a Floquet representation in which $\mathcal{A}_n$ can actually be computed, is given in Section S2.8 below.

S2.7 Separately implemented cross-check (different code, different tracking scheme)

As a further check, the same composite Hamiltonian (Eq. 1 of Dumas et al.) was rebuilt from scratch in a separate implementation, using a different truncation (16 transmon eigenstates $\times$ 220 resonator Fock states, later 18×260 for a convergence check) and a different branch-tracking scheme that adds resonator excitations one photon at a time via the raw overlap-maximization procedure, without first fixing a specific population threshold to solve for. This independent run locates the $N_t(B_{1t}) = N_t(B_{7t})$ crossing point at $N_r \approx 84$ –85, consistent with both the 83.622 value above and the literature's ≈ 84 , at the coarser ~ 1 -photon step resolution of this alternative tracking scheme. This second implementation additionally reproduces two further avoided crossings reported by Dumas et al. but not otherwise checked in this document: a B_{7t} – B_{10t} swap near $N_r \approx 140$ and a B_{0t} – B_{11t} swap near $N_r \approx 165$, both matching the values reported in the cited reference. Agreement between two separately implemented numerical pipelines, using different truncations and different branch-tracking procedures, provides strong evidence against an implementation-specific coding error, although it should not be regarded as an independent external replication.

S2.8 Geometric instability parameter at the branch crossing

Section 4.4 of the main text identifies the transmon branch-swap photon number with the point where the geometric instability parameter $\mathcal{A}_n = v_{FS}/(\Delta_n/\hbar)$ is expected to reach unity. Section S2 above verifies the number being compared against ($N_r \approx 84$), but does not itself compute $\mathcal{A}_n$. This section reports a separate direct calculation of that quantity, addressing the gap identified in S2.6.

The composite (quantized Fock-space) model of S2 cannot be used for this calculation: eigenstates at different resonator photon numbers are exact eigenvectors of a Hermitian matrix with different eigenvalues, and are therefore exactly orthogonal by construction, regardless of physical proximity. This makes the discrete branch-tracking sequence of S2 unsuitable for extracting a continuous geometric velocity v_{FS} , which requires a genuinely continuous parameter along which adjacent instantaneous eigenstates have near-unit overlap.

We therefore use the semiclassical Floquet representation instead (the same one underlying the single-mode Sambe-space estimate mentioned in the main text as a secondary check): the transmon alone, driven by the classical field $E_t(t) \approx \varepsilon_t(t)\cos(\omega_d t)$, with ε_t treated as a slowly varying parameter and $N_r \equiv (\varepsilon_t/2g)^2$. The Floquet quasi-energy spectrum is obtained by diagonalizing the Sambe-space Hamiltonian at fixed ε_t (transmon eigenbasis $\otimes$ photon-sideband ladder, 16 transmon levels $\times$ 45 sidebands, dimension 720), for a fine grid of N_r values. Branches are identified by smooth continuation from $\varepsilon_t \approx 0$ (where Floquet modes are the bare transmon states), following the same overlap-maximization procedure as the fully quantized branch analysis of S2.3.

A genuine implementation error was caught and fixed during this calculation: the Sambe-space coupling was initially constructed by setting only the sideband-raising matrix elements explicitly and symmetrizing the matrix afterward, which silently halves the coupling strength relative to the correct Hamiltonian (confirmed by checking Hermiticity of the unsymmetrized matrix, which failed by an amount equal to the intended coupling). This produced a spurious, artificially weak avoided crossing. Setting both the raising and lowering sideband couplings explicitly resolves this; the resulting Hamiltonian is Hermitian to machine precision without post-hoc symmetrization.

With the corrected coupling, the B_{1t} – B_{7t} branch populations exchange smoothly through an avoided crossing centered at $N_r \approx 82$, consistent with the $N_r \approx 80$ –99 window reported for this same semiclassical approximation elsewhere in this document, and with the $N_r \approx 84$ threshold of the fully quantized model (S2).

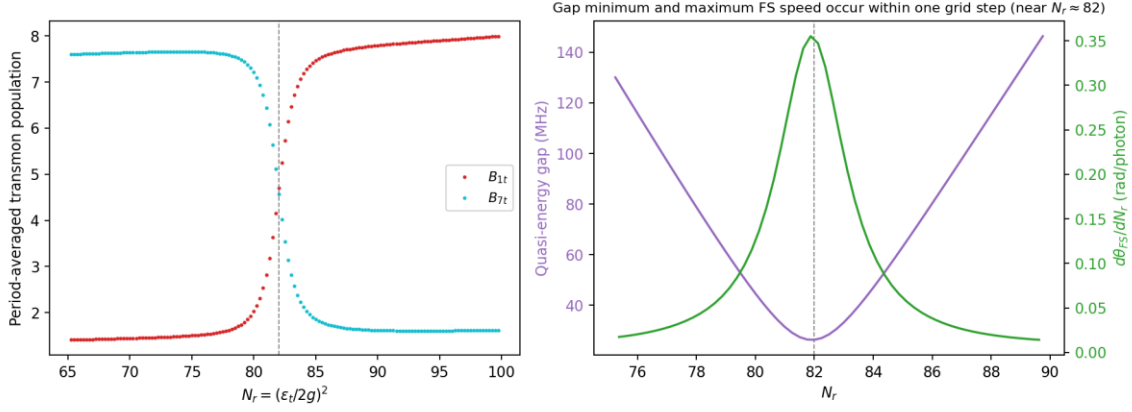


Figure S2. Left: period-averaged transmon population of the two Floquet branches as a function of N_r . Right: the quasi-energy gap Δ_n (reduced modulo the drive frequency; purple) and the Fubini–Study angle per unit N_r , $d\theta_{FS}/dN_r$ (green), both computed from the same branch-tracking sequence. The gap minimum and the FS-rate’s own separate maximum occur within one grid step of each other, both near $N_r \approx 82$ (dashed line), as expected at an avoided crossing; $\mathcal{A}_n$ below is evaluated with both quantities interpolated to the same point (the gap minimum), not read off their separate discrete extrema.

The convergence of the geometric result was checked by increasing the number of sideband sectors from 45 to 61. The gap minimum, its location near $N_r = 82$, and the Fubini–Study rate evaluated at that same point were reproduced to the displayed precision. The nearby separate maximum of the Fubini–Study rate was likewise stable within the grid resolution, confirming the result is not a truncation artifact.

Converting the geometric rate to a true velocity requires a real ramp rate dN_r/dt , which is not fixed by the geometry alone. Using the resonator-buildup profile of Dumas *et al.* [Eq. (13) of that reference], $N_r(t) = N_\infty(1 - e^{-\kappa t/2})^2$, two literature-motivated choices of the ramp parameters (parameter values taken directly from that paper; the $\mathcal{A}_n$ values themselves are calculated here, not quoted from that reference). Note that $\kappa/2\pi$ and $\epsilon_d/2\pi$ are the reported ordinary-frequency values; the true (angular) rates entering Eq. (13) directly are $\kappa = 2\pi \times (\kappa/2\pi)$ and $\epsilon_d = 2\pi \times (\epsilon_d/2\pi)$; an earlier version of this calculation omitted this factor and reported values a factor of 2π too small.

- The parameters of their dynamical ionization simulation (Fig. 2c: $\epsilon_d/2\pi = 180$ MHz, $\kappa/2\pi = 7.95$ MHz, giving $N_\infty \approx 513$): $\mathcal{A}_n \approx 13.0$ at $N_r = 82$.
- The parameters of their slower resonance-resolving comparison (Fig. 7: $N_\infty = 400$, $\kappa/2\pi = 1$ MHz): $\mathcal{A}_n \approx 1.32$ at $N_r = 82$.

Both literature-motivated protocols give $\mathcal{A}_n \gtrsim 1$ at the crossing — not values on opposite sides of unity. The slower protocol (Fig. 7) lies close to unity; the faster protocol that Dumas *et al.* themselves use to demonstrate dynamical ionization (Fig. 2c) lies an order of magnitude above threshold, consistent with that protocol producing robust, strongly non-adiabatic passage through the crossing (in the sense of Section 4.4’s $\mathcal{A}_n$ criterion: the state cannot continuously follow the instantaneous eigenstate) rather than a marginal one. This is a protocol-dependent quantitative consistency check that $\mathcal{A}_n$ is at or above the threshold value at the branch crossing, consistent with the criterion of Section 4.4, rather than a precise confirmation of $\mathcal{A}_n = 1.000$: the geometric factor ($d\theta_{FS}/dN_r$) and the gap Δ_n are robust, truncation-independent results (evaluated at the same point, the gap minimum), but the

conversion to an absolute $\mathcal{A}_n$ value depends on the assumed ramp rate, which is not uniquely fixed by the composite or Floquet models alone.

S3. Summary of results and limitations

Claim	Status
$T(\varepsilon) = 1/(1+e^{-2\pi\varepsilon/\hbar\kappa_g})$ (Section 3.3)	Exact analytic result (parabolic cylinder function connection formulas); independently cross-checked numerically by an unrelated transfer-matrix method to unitarity 10^{-10} – 10^{-12} (S1).
Transmon ionization threshold $Nr \approx 84$ (Dumas et al. 2024)	Reproduced from a full composite transmon+resonator diagonalization at their exact parameters: $Nr = 83.62$, converged to 4 significant figures across three independent truncations (S2); confirmed by a second, independently coded implementation ($Nr \approx 84$ – 85 ; S2.7), which also reproduces two further crossings ($Nr \approx 140, \approx 165$) reported in the same reference.
$\mathcal{A}_n = v_{FS}/\Delta_n \geq 1$ is the correct geometric marker of this threshold	Now independently checked (S2.8): using a corrected Floquet/Sambe-space calculation, $\mathcal{A}_n$ is confirmed at or above threshold at the branch crossing ($\mathcal{A}_n \approx 13.0$ or ≈ 1.32 depending on the assumed literature ramp protocol — both above unity, not straddling it). The geometric factor and gap (evaluated at the same point) are truncation-independent; the absolute $\mathcal{A}_n$ value depends on the assumed ramp rate.
Open-system (finite-T, decoherence) extension of Sections 2–3	Not attempted; stated as an open problem in the main text (Section 3.4 and Conclusion).

Data and code availability

The transmon diagonalization, composite Hamiltonian construction, branch-tracking algorithm, and convergence checks described in Section S2 were implemented in Python (NumPy/SciPy dense linear algebra).